QUESTION-TO-SLIDE MAPPING

SECTION A - LECTURE 11

Non-Thermal Microclimate in Designed Space

Course	BECM 3115 - Climate and Architectural Design
Lecture deck	11.0 - Non thermal Microclimate in Design Space (10 PDF pages/slides)
Question bank	Regular term-final examinations, 2016-2024 (8 supplied papers; 2019 not present)
Mapping rule	Map ventilation, air-space, infiltration and numerical questions only to what is shown in Lecture 11. The lecture table is treated as the authoritative course table. Where scanned exam tables reorder headings, the report flags the discrepancy rather than silently adopting the scan order.

RESULT	Lecture 11 is the strongest recent numerical hot zone in Section A. Air-space/heat-loss, fire-shelter and laboratory suitability questions repeat in 2021, 2022, 2023 and 2024, while room-ventilation numericals repeat in 2021-2023. Slides 2-9 directly support the entire family.

Prepared from the uploaded lecture deck and the supplied 2016-2024 regular final question bank.

1. Executive Summary

Lecture 11 defines ventilation/air change, explains how ventilation rate is expressed, defines air-space and infiltration rate, gives the course fresh-air table, and then supplies five numerical problem types. The recent question bank closely mirrors these slide problems, making the lecture unusually predictable. The safest strategy is to memorize the table, units and workflow rather than isolated answers.

10	17	17	0	0	130
Slides reviewed	Question links	Exact matches	Direct matches	Partial matches	Mapped marks

Highest-priority slide blocks

Slides	Lecture block	Historical signals	Exam value
2-3	Ventilation definition, objectives, flow-rate and ACH expressions	2021 Q5(a); 2022 Q8(d)	Direct short-note/definition marks and numerical setup.
4-6	Air-space, infiltration, fresh-air table	2021-2024 air-space/shelter/lab family	Core data block; must be memorized/read correctly.
7	Room ventilation + heat-loss example types	2021-2024 repeated numericals	Highest-value numerical setup.
8	Laboratory + fire-shelter problem types	2021-2024 four consecutive papers	Near-direct templates for recent exam questions.
9	Occupancy from infiltration rate	Related ventilation capacity logic	Useful variation; less visible as a separate recent final question.

Priority note Memorize the course table at 3/6/9/12 m3 per person and the three columns Minimum / Smoking / Non-smoking. Then practice one room-ventilation, one heat-loss, one shelter and one laboratory problem without notes.

2. Detailed Year-wise Question Mapping

The original printed section label may change from year to year. Mapping below follows subject matter and the lecture content. Slide numbering uses the PDF page count, including the title slide.

Year	Original	Marks	QB p.	Question / task	Match	Lecture slides	Coverage assessment
2021	Q5(a)	5	6	What does air change mean? How is ventilation rate expressed in a built space?	Exact	2-3	The definition/objectives and both rate-expression methods are directly stated.
2021	Q5(d)	8	6	Room 45 x 23 x 3 m, 200 people, smoking prohibited: find m3/s, L/s and air changes/hour.	Exact	3-7	The lecture gives the rate forms, air-space/table logic and an almost identical problem template.
2021	Q6(a)	10	7	Define air space and infiltration rate; find heat loss due to air change.	Exact	4-7	Definitions, table and heat-loss problem type are in the lecture.
2021	Q7(a)	10	7	Fire-shelter safety: check air-space/fresh-air adequacy and suggest improvement.	Exact	4-8	Slide 8 supplies the same shelter-appropriateness workflow and asks for improvement.
2021	Q8(c)	5	7	Laboratory fresh-air suitability for users.	Exact	4-8	Slide 8 provides the laboratory microclimate appropriateness problem type.
2022	Q5(d)	8	8	Room 45 x 23 x 3 m, 150 people, smoking prohibited: find ventilation and ACH.	Exact	3-7	Direct numerical variant of the lecture template.
2022	Q6(a)	10	9	Air space + infiltration rate + heat loss due to air change.	Exact	4-7	Same concepts and problem type as Lecture 11.
2022	Q7(d)	10	9	Fire-shelter microclimate safety for increased occupants; suggest improvement.	Exact	4-8	Direct shelter-template repeat.
2022	Q8(c)	5	9	Laboratory ventilation suitability for 20 users.	Exact	4-8	Direct lab-template repeat.

2. Detailed Year-wise Question Mapping - continued

Year	Original	Marks	QB p.	Question / task	Match	Lecture slides	Coverage assessment
2022	Q8(d)	6	9	Short notes: air space, air change and infiltration rate.	Exact	2-5	All three terms are directly covered.
2023	Q5(d)	10	11	Air space/infiltration rate + heat loss due to air change.	Exact	4-7	Same repeated template.
2023	Q6(d)	7	11	Room 45 x 25 x 3 m, 180 people: find required ventilation and ACH.	Exact	3-7	Direct room-ventilation variant.
2023	Q7(c)	10	12	Fire-shelter safety and improvement.	Exact	4-8	Direct repeated shelter template.
2023	Q8(b)	5	12	Laboratory fresh-air suitability / power-failure condition.	Exact	4-8	Lecture gives the same lab suitability logic; power failure is an application of losing forced supply.
2024	Q5(d)	5	14	Laboratory ventilation suitability for 20 users, normal and power failure.	Exact	4-8	Direct recent repetition of the lecture lab problem family.
2024	Q6(d)	8	14	Fire-shelter safety: volume, occupant load and available fresh air; suggest improvement.	Exact	4-8	Direct shelter-template repeat.
2024	Q7(a)	8	14	Air space/infiltration rate + heat loss due to air change.	Exact	4-7	Direct repeated heat-loss family.

Coverage key	Exact = answer content is essentially present as asked. Direct = most content is present, but the student must apply/combine it. Partial = only one component is present; another lecture/source is required.

3. Repeated Question Patterns

Recent papers reuse the same numerical logic with changed dimensions, occupants and air-supply values. The safest preparation is therefore a fixed workflow, not memorization of any one year's numbers.

P	Normalized repeated family	Years seen	Slides	Why it matters
S	Air-space/infiltration + heat loss due to air change	2021, 2022, 2023, 2024	4-7	Four consecutive papers; combines definition + formula workflow.
S	Fire-shelter capacity / safety / improvement	2021, 2022, 2023, 2024	4-8	Four consecutive papers with nearly identical logic.
S	Laboratory ventilation suitability / power failure	2021, 2022, 2023, 2024	4-8	Four consecutive papers; easy if table selection is correct.
A	Room ventilation in m3/s, L/s and ACH	2021, 2022, 2023	3-7	Three consecutive papers; unit conversion + table + ACH.
A	Air change / air space / infiltration short notes	2021, 2022, 2024 related	2-5	Definitions are direct and low-risk marks.

4. Quick Recognition Rules

If the question says...	Go to slides	Recognition rule
“air change / ventilation rate expressed”	2-3	State fresh-air purpose, then flow quantity (m3/s or L/s) and air changes/hour.
“air space per person”	4	Net built-space volume divided by number of users; subtract men/furniture only if the problem requires it.
“fresh-air requirement / smoking / non-smoking”	5-6	Choose air-space row first, then the correct column; multiply by people.
“fire shelter / laboratory suitable?”	4-8	Check BOTH air-space/person and available fresh-air supply; then give a safety verdict/improvement.
“heat loss due to air change”	7	Convert ACH to volumetric flow, then use the heat-loss relation supplied/used in class; state units and any extra assumption.

5. Slide-to-Question Reverse Index

Use this page while revising the lecture: start from a slide block, then immediately practice the linked past questions.

Slides	What the slides teach	Past-question links	Study action
2	Ventilation definition + five objectives	2021 Q5(a); 2022 Q8(d)	Memorize as one 5-point list.
3	Ventilation rate: m3/s or L/s; air changes/hour	2021 Q5(a); 2021-23 room numericals	Write units before calculation.
4	Air-space definition and formula	2021-24 air-space/shelter/lab	Practice net volume/person calculation.
5-6	Infiltration + course fresh-air table	All recent numerical families	Memorize table and column order.
7	Problem 1 room ventilation; Problem 2 heat loss	2021-24	Practice both problem types from blank paper.
8	Problem 3 lab; Problem 4 fire shelter	2021-24	These are the direct templates for the modern repeated trio.
9	Problem 5 occupants from infiltration	Related capacity problems	Use as a reverse-calculation exercise.
10	Closing	None	Skip.

6. Coverage Gaps & Diagnostic Exclusions

Lecture 11 is unusually complete for the final-exam numerical family, but a few formulas/assumptions used in full solutions are not printed explicitly on the slides. Those should not be misrepresented as slide content.

Gap	Affected questions	Action
General ACH algebraic formula is not printed as an equation	Room-ventilation numericals	The slide states both rate forms; use the class-note formula if required, but label it as derived/class-note support rather than slide text.
Air density is not supplied on the heat-loss slide	2021-2024 heat-loss numericals	If the paper/class convention does not give density, state the assumption used rather than pretending it came from Lecture 11.
AH-SH-RH relation is not part of Lecture 11	2020/2021 relative-humidity questions	Use Lecture 06 / comfort-humidity material; do not map RH numericals here.
Cross-ventilation opening arrangement is not developed here	2018/2023 airflow-layout questions	Use Lecture 07; Lecture 11 focuses on quantity/air-space rather than architectural airflow geometry.

Mapping discipline The lecture table consistently gives, at 3 m3/person, Minimum = 11.3 L/s, Smoking = 22.6 L/s, Non-smoking = 17.0 L/s. Some scanned final-paper tables reorder headings. This report follows the lecture deck and flags the scan inconsistency instead of silently changing the course table.

7. Recommended Study Plan

Use a fixed four-step numerical routine: Volume -> Air space/person -> Table rate -> Total flow / ACH / verdict. For heat loss, add the heat-transfer step only after the airflow quantity is correct.

Step	Study block	Slides	Practice immediately
1	Definitions + units + table	2-6	2021 Q5(a), 2022 Q8(d).
2	Room ventilation / ACH	7	2021 Q5(d), 2022 Q5(d), 2023 Q6(d).
3	Air-change heat loss	7	2021 Q6(a), 2022 Q6(a), 2023 Q5(d), 2024 Q7(a).
4	Fire-shelter safety	8	2021 Q7(a), 2022 Q7(d), 2023 Q7(c), 2024 Q6(d).
5	Laboratory suitability	8	2021 Q8(c), 2022 Q8(c), 2023 Q8(b), 2024 Q5(d).

Exam-ready minimum

Priority	What to reproduce	Exam technique
A	Fresh-air table	Write row/column headings correctly before substituting numbers.
A	Unit conversion chain	L/s -> m3/s before ACH; show units at every line.
A	Safety verdict	Do not stop at arithmetic: state suitable/unsuitable and why.
A	Improvement list	Increase fresh-air supply / increase volume / reduce occupancy / provide reliable backup ventilation.

Bottom line	Lecture 11 is one of the safest modern-paper scoring areas. The wording changes, but the numerical skeleton barely changes. Master the table and one workflow, and four years of repeated questions become manageable.